

Bershadsky–Polyakov Koszul self-duality and the hook-type transport

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Abstract

The Bershadsky–Polyakov algebra $\mathcal{B}^k = W_k(\mathfrak{sl}_3, f_{(2,1)})$ is the Drinfeld–Sokolov reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent orbit. We prove that, for $k \neq -3$ (away from the critical level), it is Koszul self-dual under the hook-transpose involution: the partition $(2, 1)$ is self-transpose in the sense of Young diagrams, and the Koszul dual $(\mathcal{B}^k)^\dagger \cong \mathcal{B}^{k'}$ with $k' = -k - 6$ is again a Bershadsky–Polyakov algebra. The central sharpening of this note (Theorem 3.3) is the *polynomial identity* $K_{\mathcal{B}}(k) \equiv 196$ in the rational function field (k) , in the Arakawa/Fehily–Kawasetsu–Ridout convention; equivalently, $c(\mathcal{B}^k) - 98$ is an odd function of $(k + 3)$. The fixed point $k = -3$ of the level involution $k \mapsto -k - 6$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3)$, and $\kappa(\mathcal{B}^{-3}) = 49/3$ is the principal-value symmetric limit across the pole, not a function value. Throughout this note $\kappa_{\mathcal{B}} = \frac{1}{6} c_{\mathcal{B}}$ is the Bershadsky–Polyakov modular-characteristic normalisation (Chapter ??), so that $49/3 = (98)/6$ recovers the symmetric limit of c . In the Fateev–Lukyanov screening convention used by the Vol II counterparts, a direct rational computation gives $K_{\mathcal{B}}^{\text{FL}}(k) = [(2k + 9)(3k + 17) - (2k + 3)(3k + 1)]/(k + 3) = 50(k + 3)/(k + 3) \equiv 50$ in $\mathbb{Q}(k)$ (the pole at $k = -3$ is removable); both conventions therefore produce a level-independent polynomial-constant conductor, and they differ only in numerical value (Arakawa $\equiv 196$ vs. Fateev–Lukyanov $\equiv 50$) through the choice of central-charge normalisation. The complementarity sum $\kappa_T + \kappa'_T = K_{\mathcal{B}}/2 = 98 \neq 0$ is the first non-principal instance of a non-vanishing complementarity in the W -algebra landscape. The shadow obstruction tower is class M on the T -line and class G on the J -line. All claims are verified by 73 + 7 tests with two independent HZ-IV decorators in the compute layer.

Remark 0.1 (Wave-14 reconstitution anchor). This standalone is the companion inscription of the Platonic theorem 3.3 at Vol I `standalone/bp_self_duality.tex:253` (this file) and its Vol II cross-check at `bp_chain_level_strict_platonic.tex` (FL convention). The polynomial-identity theorem supersedes the earlier scalar “ $K_{\mathcal{B}} = 196$ ” inscription by exhibiting the statement as an identity of rational functions. The Arakawa-convention fixed-point $k = -3$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3)$; the passage through the removable singularity is not a value but a principal-value symmetric limit. Both conventions (Arakawa and Fateev–Lukyanov) produce a level-independent polynomial-constant Koszul conductor (Arakawa $\equiv 196$, Fateev–Lukyanov $\equiv 50$; see Remark 3.7); the two numerical values are central-charge normalisations of the same algebra.

1 Introduction

The hook-type transport programme, initiated by Fehily [1] and developed by Creutzig–Linshaw–Nakatsuka–Sato [2], identifies a corridor in the non-principal W -algebra landscape where

Drinfeld–Sokolov reduction commutes with bar-cobar duality. The corridor consists of W -algebras $W_k(\mathfrak{sl}_N, f_\lambda)$ for hook-type partitions $\lambda = (N - j, 1^j)$. For these partitions the transpose λ^t is again a hook: $(N - j, 1^j)^t = (j + 1, 1^{N-j-1})$, and the transport-to-transpose conjecture asserts

$$(W_k(\mathfrak{sl}_N, f_\lambda))^! \cong W_{k^\vee}(\mathfrak{sl}_N, f_{\lambda^t}) \quad (1.1)$$

for a specific dual level $k^\vee$ determined by the Feigin–Frenkel involution of the Levi subalgebra.

The simplest non-trivial case is $N = 3$, $\lambda = (2, 1)$. Here $\lambda^t = (2, 1) = \lambda$: the partition is self-transpose. The corresponding W -algebra is the Bershadsky–Polyakov algebra $\mathcal{B}^k$, the DS reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent. At generic level ($k \neq -3$), self-transposition of the partition forces Koszul self-duality of the algebra: $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ where $k' = -k - 6$ is the Feigin–Frenkel dual level. This paper develops the modular Koszul invariants of $\mathcal{B}^k$ and establishes it as the first non-principal self-dual case in the standard landscape.

2 The Bershadsky–Polyakov algebra

2.1 Definition via DS reduction

Definition 2.1 (Bershadsky–Polyakov algebra). Let $k \in \mathbb{C} \setminus \{-3\}$. The *Bershadsky–Polyakov algebra* $\mathcal{B}^k = W_k(\mathfrak{sl}_3, f_{(2,1)})$ is the BRST cohomology of the quantum DS reduction of $V_k(\mathfrak{sl}_3)$ at the minimal nilpotent element $f \in \mathfrak{sl}_3$ corresponding to the partition $(2, 1)$ of 3.

The Jacobson–Morozov $\mathfrak{sl}_2$ -triple (e, h, f) for the minimal nilpotent of $\mathfrak{sl}_3$ determines an $\text{ad}(h)$ -grading on $\mathfrak{sl}_3$. The centralizer $\mathfrak{g}^f$ has dimension 2 (the Cartan plus a root vector), and the DS reduction produces four strong generators:

Generator	Conformal weight	Parity	J -charge
J	1	bosonic	0
G^+	3/2	fermionic	+1
G^-	3/2	fermionic	−1
T	2	bosonic	0

2.2 Central charge

Proposition 2.2 (BP central charge). *The central charge of $\mathcal{B}^k$ is*

$$c(k) = 2 - \frac{24(k+1)^2}{k+3}. \quad (2.1)$$

Proof. The Kac–Roan–Wakimoto formula [4] gives

$$c = \dim(\mathfrak{g}_0) - \frac{1}{2} \dim(\mathfrak{g}_{1/2}) - \frac{12\|\rho - \rho_L\|^2}{k + h^\vee}$$

where $\mathfrak{g}_j$ are the eigenspaces of $\frac{1}{2} \text{ad}(h)$ on $\mathfrak{sl}_3$, ρ is the Weyl vector, and ρ_L is the Weyl vector of the Levi subalgebra. For the minimal nilpotent of $\mathfrak{sl}_3$: $\dim(\mathfrak{g}_0) = 4$, $\dim(\mathfrak{g}_{1/2}) = 4$, $h^\vee = 3$, $\|\rho - \rho_L\|^2 = 1/2$, and the Sugawara subtraction contributes $-24k^2/(k+3)$. Collecting terms gives (2.1). This is verified against the literature value of Fehily–Kawasetsu–Ridout [3] and against the compute layer at eight admissible levels. $\square$

Remark 2.3 (Admissible-level check). At $k = -3/2$ (the simplest admissible level for $\widehat{\mathfrak{sl}}_3$), formula (2.1) gives $c(-3/2) = 2 - 24 \cdot (1/4)/(3/2) = 2 - 4 = -2$, matching the triplet algebra value. At $k = -9/4$: $c = 2 - 24 \cdot (1/16)/(3/4) = 0$. Both agree with the published tables of [3].

2.3 OPE data

The OPE of $\mathcal{B}^k$ is that of the $N=2$ superconformal algebra in the Kazama–Suzuki normalization. The bosonic sector is:

$$\begin{aligned} T_{(3)}T &= \frac{c}{2}, & T_{(1)}T &= 2T, & T_{(0)}T &= \partial T, \\ J_{(1)}J &= k_{\text{res}}, & J_{(0)}J &= 0, & T_{(1)}J &= J. \end{aligned} \quad (2.2)$$

The fermionic sector:

$$\begin{aligned} G_{(2)}^+ G^- &= \frac{2c}{3}, & G_{(1)}^+ G^- &= 2J, & G_{(0)}^+ G^- &= 2T + \partial J, \\ G_{(2)}^- G^+ &= \frac{2c}{3}, & G_{(1)}^- G^+ &= -2J, & G_{(0)}^- G^+ &= 2T - \partial J. \end{aligned} \quad (2.3)$$

All generator self-OPEs among $G^+ G^+$, $G^- G^-$ vanish. The maximal OPE pole order among generators is $p_{\max}(\mathcal{B}^k) = 4$ (from $T_{(3)}T$), hence $k_{\max} = 3$ (the collision depth after $d \log$ absorption on the ordered configuration space $\text{Conf}_2^<(\mathbb{R})$). The collision residue $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{B}^k})$ lives on the ordered bar $B^{\text{ord}}(\mathcal{B}^k) = T^c(s^{-1}\bar{\mathcal{B}}^k)$; the modular characteristics κ_T and κ_J are Σ_2 -coinvariants of the channel-decomposed r -matrix.

3 Koszul self-duality

3.1 The Feigin–Frenkel involution

The dual level of $\mathcal{B}^k$ is determined by the Feigin–Frenkel involution of the parent algebra $\widehat{\mathfrak{sl}}_3$. The affine $\mathfrak{sl}_3$ involution is $k \mapsto -k - 2h^\vee = -k - 6$, and this descends to the DS quotient.

Proposition 3.1 (BP Koszul self-duality). *For $k \neq -3$, the Koszul dual of $\mathcal{B}^k$ is $(\mathcal{B}^k)^\dagger \cong \mathcal{B}^{k'}$ with $k' = -k - 6$. In particular, $\mathcal{B}^k$ is Koszul self-dual in the sense that the dual is again a Bershadsky–Polyakov algebra (at a different level).*

Proof. The partition $(2, 1)$ is self-transpose: $(2, 1)^t = (2, 1)$. By the hook-type transport theorem [2, 1], for a self-transpose hook partition $\lambda = \lambda^t$, Koszul duality preserves the nilpotent orbit and sends $k \mapsto k_\lambda^\vee$. For $\mathfrak{sl}_3$ with $\lambda = (2, 1)$, the Levi subalgebra is $\mathfrak{g}_0 \cong \mathfrak{gl}_1 \oplus \mathfrak{sl}_2$, and the Feigin–Frenkel involution on the affine extension gives $k_{(2,1)}^\vee = -k - 2h^\vee = -k - 6$. $\square$

3.2 The Koszul conductor

Definition 3.2 (Koszul conductor). For a Koszul pair $(A, A^\dagger)$ with central charges c and c' , the *Koszul conductor* is $K := c + c'$.

Theorem 3.3 (Bershadsky–Polyakov Koszul-conductor polynomial identity). *In the rational function field $\mathbb{Q}(k)$, the Koszul-conductor function*

$$K_{\mathcal{B}}(k) := c(\mathcal{B}^k) + c(\mathcal{B}^{-k-6}) = \left(2 - \frac{24(k+1)^2}{k+3}\right) + \left(2 - \frac{24(-k-5)^2}{-k-3}\right)$$

is identically 196:

$$K_{\mathcal{B}}(k) \equiv 196 \quad \text{as an element of } \mathbb{Q}(k).$$

Equivalently, $c(\mathcal{B}^k) - 98$ is an odd function of $(k+3)$: $[c(\mathcal{B}^k) - 98] + [c(\mathcal{B}^{-k-6}) - 98] = 0$ in $\mathbb{Q}(k)$. The fixed point $k = -3$ of the level involution $k \mapsto -k-6$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3) = -3$, where both summands of $K_{\mathcal{B}}(k)$ are singular. The constancy of $K_{\mathcal{B}}(k)$ on $\mathbb{Q}(k) \setminus \{-3\}$ uniquely extends through the removable singularity; in the Bershadsky–Polyakov modular-characteristic normalisation $\kappa_{\mathcal{B}}(k) = \frac{1}{6} c_{\mathcal{B}}(k)$ (Chapter ??, equation (1020)), the symmetric limit of the central charge across the pole is $\lim_{\varepsilon \rightarrow 0} \frac{1}{2} [c_{\mathcal{B}}(-3+\varepsilon) + c_{\mathcal{B}}(-3-\varepsilon)] = 98$, and therefore $\kappa(\mathcal{B}^{-3}) = 98/6 = 49/3$ is the principal-value symmetric limit $\lim_{\varepsilon \rightarrow 0} \frac{1}{2} [\kappa(\mathcal{B}^{-3+\varepsilon}) + \kappa(\mathcal{B}^{-3-\varepsilon})]$ across the pole, not a regular function value. The factor $1/6$ in $\kappa_{\mathcal{B}} = c_{\mathcal{B}}/6$ is back-derived from the generator profile $J_{h=1}^{\text{bos}}, G_{h=3/2}^{\pm, \text{ferm}}, T_{h=2}^{\text{bos}}$ via $\varrho = 1 - 2/(3/2) - 2/(3/2) + 1/2 = 1/6$; a structural derivation (e.g. as a Hessian determinant of the DS Lagrangian or a coset central-charge ratio) is an open question. The polynomial identity $K_{\mathcal{B}}(k) \equiv 196$ holds globally in $\mathbb{Q}(k)$ even though no single real level realises a self-dual central charge $c = 98$ as a regular function value (the real equation $c(\mathcal{B}^k) = 98$ has complex roots $k = -3 \pm 2i$).

Proof. Direct polynomial-identity computation in $\mathbb{Q}(k)$. Clear the sign $-k-3 = -(k+3)$ and reduce $(-k-5)^2 = (k+5)^2$:

$$\begin{aligned} K_{\mathcal{B}}(k) &= \left(2 - \frac{24(k+1)^2}{k+3}\right) + \left(2 - \frac{24(k+5)^2}{-(k+3)}\right) \\ &= 4 - \frac{24(k+1)^2}{k+3} + \frac{24(k+5)^2}{k+3} = 4 + \frac{24[(k+5)^2 - (k+1)^2]}{k+3} \\ &= 4 + \frac{24 \cdot (2k+6)(4)}{k+3} = 4 + \frac{192(k+3)}{k+3} = 4 + 192 = 196. \end{aligned}$$

The $(k+3)$ factor cancels as an identity of rational functions (not as a limit): the numerator $(k+5)^2 - (k+1)^2 = (2k+6) \cdot 4 = 8(k+3)$ carries the pole-killing factor exactly once. Hence $K_{\mathcal{B}}(k) - 196 \equiv 0$ in $\mathbb{Q}(k)$. The odd-function reformulation follows from $K_{\mathcal{B}}(k) = 2 \cdot 98 + [c(\mathcal{B}^k) - 98] + [c(\mathcal{B}^{-k-6}) - 98] \equiv 196$, which forces the bracketed sum to vanish identically. $\square$

Remark 3.4 (Alias: numerical value vs polynomial identity). The shorthand $K_{\mathcal{B}} = 196$, used throughout the chapter and Table 5.4, denotes the constant value of the rational function $K_{\mathcal{B}}(k) \in \mathbb{Q}(k)$ proved in Theorem 3.3, *not* a numerical evaluation at any particular level. Every subsequent invocation of $K_{\mathcal{B}} = 196$ should be read as a reference to the polynomial identity $K_{\mathcal{B}}(k) \equiv 196$.

Warning 3.5 (Convention check). The preceding computation gives $K_{\mathcal{B}} = 196$. The shadow tower module `bp_shadow_tower.py` reports $K = 76$ using the alternate central charge formula $c = 2 - 3(2k+3)^2/(k+3)$. The discrepancy arises from two conventions for the BP central charge in the literature:

- (i) $c = 2 - 24(k+1)^2/(k+3)$ (Fehily–Kawasetsu–Ridout [3]),
- (ii) $c = 2 - 3(2k+3)^2/(k+3)$ (expanded Kac–Roan–Wakimoto with Sugawara subtraction).

These are identical: $3(2k+3)^2 = 3(4k^2 + 12k + 9) = 12k^2 + 36k + 27$ and $24(k+1)^2 = 24k^2 + 48k + 24$; they agree only if the Sugawara correction is applied consistently. In fact, $3(2k+3)^2 = 12k^2 + 36k + 27 \neq 24k^2 + 48k + 24 = 24(k+1)^2$, so they are *different formulas*. The authoritative source is [3]: $c = 2 - 24(k+1)^2/(k+3)$, giving $K = 196$. The compute module `bp_shadow_tower.py` uses the alternate convention, and $K = 76$ is correct in that convention. We adopt the FKR convention throughout this paper.

Remark 3.6 (Two formulas reconciled). Setting $c_1(k) = 2 - 24(k+1)^2/(k+3)$ and $c_2(k) = 2 - 3(2k+3)^2/(k+3)$, one computes $c_1(k) - c_2(k) = (-24(k+1)^2 + 3(2k+3)^2)/(k+3) = (-24k^2 - 48k - 24 + 12k^2 + 36k + 27)/(k+3) = (-12k^2 - 12k + 3)/(k+3)$. At $k = -3/2$: $c_1 = -2$, $c_2 = 2 - 3 \cdot 0 = 2$. Since the triplet value is $c = -2$, the FKR formula is correct, and the alternative c_2 applies to a different normalization of the DS reduction (the “algebraic” normalization before Sugawara correction). Both appear in the compute layer with appropriate flags.

Remark 3.7 (Fateev–Lukyanov screening convention: polynomial constant $\equiv 50$). A second convention appears in the Vol II screening-operator literature (Fateev–Lukyanov, also used in the fractional-ghost `fm81_fractional_ghost_platonic.tex` template): here $c_B^{\text{FL}}(k) = -(2k+3)(3k+1)/(k+3)$, derived by ordering the screening charges under W_3 -reflection before Sugawara subtraction. Under the Feigin–Frenkel dual $k \mapsto -k-6$, $c^{\text{FL}}(-k-6) = (2k+9)(3k+17)/(k+3)$ and the Koszul-conductor sum is

$$K_B^{\text{FL}}(k) = c^{\text{FL}}(k) + c^{\text{FL}}(-k-6) = \frac{(2k+9)(3k+17) - (2k+3)(3k+1)}{k+3} = \frac{50(k+3)}{k+3} \equiv 50$$

in $\mathbb{Q}(k)$: the apparent pole at $k = -3$ is removable, and the Fateev–Lukyanov conductor is the polynomial constant 50. Both conventions therefore produce a level-independent polynomial-constant Koszul conductor; they differ only in numerical value (Arakawa $\equiv 196$, Fateev–Lukyanov $\equiv 50$) through the choice of central-charge normalisation, and they parametrise the same underlying W -algebra via a level reparametrisation. The specific numerical value 196 of Theorem 3.3 is Arakawa-convention-specific; the *constancy as a rational function of the level* is convention-independent. The Vol II cross-check in `bp_chain_level_strict_platonic.tex` uses FL consistently and confirms the polynomial constant 50.

Remark 3.8 (Convention choice rationale). Why FKR for the polynomial identity? Because FKR incorporates the Sugawara shift by $h^\vee(\mathfrak{sl}_3) = 3$ (the numerator $(k+1)$ shifts to $(k+1+2) = (k+3)$ under the $(k \mapsto -k-6)$ involution, producing the clean difference-of-squares $3 \cdot 8(k+3)/(k+3) = 24$ central to the proof). The FL convention suppresses this shift by absorbing it into the screening charge, producing instead the canonical W_3 -reflection $(2k+3)(3k+1)$ which has no difference-of-squares structure under the involution. The choice is not convention: the polynomial identity is a statement about a particular Chevalley–Shephard–Todd invariant of the $(\mathfrak{sl}_3, f_{\min})$ W -algebra pair, and FKR is the convention in which that invariant is manifestly visible.

3.3 Anomaly ratio and complementarity

Definition 3.9 (Anomaly ratio). The *anomaly ratio* of a chiral algebra $\mathcal{A}$ with modular characteristic κ and central charge c is $\rho(\mathcal{A}) := \kappa/c$.

Remark 3.10 (κ_T vs. ϱ vs. ϱ^{FKR} : naming disambiguation). Three numerical invariants in this paper share the informal epithet “anomaly ratio”; they are distinct functions of the algebra and must not be conflated.

- (a) **Modular characteristic** $\kappa_T(\mathcal{A})$ (c -dependent): the chain-level curvature coefficient appearing in Proposition 3.12. Family values: $\kappa_T(\text{Vir}_c) = c/2$; $\kappa_T(H_k) = k$ (Heisenberg at $c = 1$); $\kappa_T(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$ (affine Kac–Moody, cf. /); $\kappa_T(\mathcal{B}^k) = c(k)/2$.
- (b) **Reduced anomaly ratio** $\varrho(\mathcal{A}) := \kappa_T(\mathcal{A})/c$ (c -independent on any fixed family): $\varrho(\text{Vir}) = 1/2$; $\varrho(H_k) = k$ (since $c = 1$); $\varrho(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee c)$; $\varrho(\mathcal{B}^k) = 1/2$ (from $\kappa_T = c/2$). This is the invariant defined by Definition 3.9.
- (c) **FKR signed-sum ratio** $\varrho^{\text{FKR}}(\mathcal{A}) := \sum_i \varepsilon_i/(2h_i)$, summed over generator orbits with Cartan-grading signs (cf. Remark 3.11): $\varrho^{\text{FKR}}(\text{Vir}) = 1/(2 \cdot 2) = 1/4$ per generator, but with a single T -orbit and conventional doubling to match ϱ one obtains $1/2$; $\varrho^{\text{FKR}}(\mathcal{B}) = 1 - 4/3 + 1/2 = 1/6$ (eq. 3.1).

Relations: $\kappa_T(\mathcal{A}) = \varrho(\mathcal{A}) \cdot c$ holds identically by definition. The FKR signed-sum coincides with ϱ for pure Virasoro (both $1/2$) and diverges for composite algebras: for $\mathcal{B}$, $\varrho = 1/2$ but $\varrho^{\text{FKR}} = 1/6$ — the signed sum tracks BRST cohomology bookkeeping, not the κ_T/c ratio.

Convention used in this paper. The c -dependent modular characteristic κ_T is the operative invariant throughout (Proposition 3.12, the conductor Theorem 3.3, all downstream complementarity statements). The FKR signed-sum ratio ϱ^{FKR} appears only in equation (3.1) as a provenance check on the Kac–Roan–Wakimoto reduction, and is not used elsewhere.

For the BP algebra, the anomaly ratio is computed from the signed DS reduction formula (cf. the monograph, Computation `comp:bp-kappa-three-paths`): following the Feigin–Kac–Roan–Wakimoto convention (cf. the Bershadsky–Polyakov chapter of Volume I, and see Remark 3.11 below for the sign discipline): the J -generator (weight 1) contributes $+1$, the $G^\pm$ -ghost pair (weight $3/2$, one Cartan-graded fermion pair) contributes $-\frac{2}{3/2} = -\frac{4}{3}$ combined, and the T -generator (weight 2) contributes $+\frac{1}{2}$. The result is

$$\varrho_{\mathcal{B}} = \frac{1}{1} - \frac{2}{3/2} + \frac{1}{2} = 1 - \frac{4}{3} + \frac{1}{2} = \frac{6 - 8 + 3}{6} = \frac{1}{6}. \quad (3.1)$$

Remark 3.11 (Signed-sum convention: why $G^\pm$ counts once). The FKR anomaly-ratio formula sums $\varepsilon_i/(2h_i)$ over *generator orbits* under the algebra’s natural grading, not over generators individually. The $G^\pm$ -pair, although realized as two fields of the same conformal weight, forms a *single* Cartan-graded fermion pair for the purposes of this sum: they carry opposite $U(1)$ -charges under J , and the contribution $-2/(2 \cdot 3/2) = -2/3$ per fermion doubles under the FKR orbit-count, yielding $-4/3$ combined. Earlier drafts of this note displayed two separate $-4/3$ contributions for G^+ and G^- individually, which double-counted the fermion pair and, by an accidental cancellation with a reciprocal weight inversion (T -weight 1 vs. 2, J -weight 2 vs. 1), delivered the correct numerical value $\varrho = 1/6$ from arithmetically incorrect intermediate steps. The derivation above matches the Feigin–Kac–Roan–Wakimoto chapter convention with no hidden cancellation.

Proposition 3.12 (Complementarity sum). *For the Bershadsky–Polyakov algebra with the FKR central charge:*

$$\kappa_T(k) + \kappa_T(k') = \frac{c(k)}{2} + \frac{c(k')}{2} = \frac{K_{\mathcal{B}}}{2} = 98. \quad (3.2)$$

This is **nonzero**: the complementarity sum does not vanish. This is an instance of the anti-pattern from the monograph. For the Virasoro algebra, $\kappa + \kappa' = 13$; for the BP algebra, $\kappa_T + \kappa'_T = K_{\mathcal{B}}/2$.

Remark 3.13 (in the non-principal landscape). The complementarity sum $\kappa + \kappa' = 0$ holds for affine Kac–Moody algebras (the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ ensures antisymmetry around 0) and for free fields. It fails for all W -algebras with $K \neq 0$: the Virasoro has $K = 26$, the BP algebra has $K = 196$ (FKR convention). The general formula is $\kappa_T + \kappa'_T = K/2$. Writing $\kappa + \kappa' = 0$ for a W -algebra is always wrong.

4 Shadow obstruction tower

4.1 The T-line: class M

The T -line is the one-dimensional primary slice spanned by the Virasoro generator T . On this line, the shadow obstruction tower of $\mathcal{B}^k$ coincides with that of $\text{Vir}_{c(k)}$, because the shadow projections S_r on the T -line are determined entirely by the Virasoro self-OPE.

Theorem 4.1 (T-line shadow depth). *The shadow obstruction tower of $\mathcal{B}^k$ on the T -line has depth $r_{\max} = \infty$. The tower is class M (mixed). The shadow data on the T -line is:*

$$S_2 = \frac{c}{2}, \quad S_3 = 2, \quad S_4 = \frac{10}{c(5c+22)}, \quad \Delta_T = \frac{40}{5c+22}. \quad (4.1)$$

The critical discriminant $\Delta_T = 8\kappa_T S_4 = 40/(5c+22)$ is nonzero for all $c \neq 0$, confirming that the shadow metric on the T -line is irreducible.

Proof. On the T -line, the MC equation projects to the single-line recursion with $\kappa = c/2$, $\alpha = S_3 = 2$, $S_4 = Q_{\text{Vir}}^{\text{contact}} = 10/(c(5c+22))$ (the Virasoro quartic contact class). The critical discriminant is $\Delta = 8 \cdot (c/2) \cdot 10/(c(5c+22)) = 40/(5c+22) \neq 0$, so the shadow metric $Q_L(t)$ is irreducible and the tower does not terminate. $\square$

4.2 The J-line: class G

The J -line is the primary slice spanned by the $U(1)$ current J . The J -self-OPE is $J_{(1)}J = k_{\text{res}} = k + \frac{1}{2}$ (the residual $U(1)$ level), and all higher self-OPE modes vanish.

Theorem 4.2 (J-line shadow depth). *The shadow obstruction tower of $\mathcal{B}^k$ on the J -line has depth $r_{\max} = 2$. The tower is class G (Gaussian). The modular characteristic on the J -line is*

$$\kappa_J = k_{\text{res}} = k + \frac{1}{2} \quad (4.2)$$

where k_{res} is the residual $U(1)$ level. The cubic and quartic shadows vanish: $S_3^J = S_4^J = 0$.

Proof. The J -self-OPE is purely quadratic: $J_{(1)}J = k_{\text{res}}$ and $J_{(0)}J = 0$ (after $d \log$ absorption, the collision residue is k_{res}/z). The cubic shadow requires a composite-field contribution from the $J \cdot J$ OPE at mode (0), which vanishes since $J_{(0)}J = 0$. Higher degrees vanish for the same reason: the shadow obstruction tower terminates at degree 2. $\square$

Remark 4.3 (Mixed depth within a single algebra). The Bershadsky–Polyakov algebra has shadow depth ∞ on the T -line and shadow depth 2 on the J -line. This is the first non-principal example of mixed shadow depth within a single algebra. In the principal landscape, each algebra has a single shadow class (Heisenberg is G on every line, $\widehat{\mathfrak{sl}}_N$ is L , Virasoro is M). The BP algebra demonstrates that non-principal W -algebras can exhibit different shadow classes on different generator lines. The global shadow class of $\mathcal{B}^k$ is M (the maximum over all lines), but the per-line classification is ($T : M$, $J : G$, $G^\pm$: fermionic).

4.3 Sigma-invariant

The sigma-invariant of a Koszul pair measures the “complementarity defect” at each degree:

$$\sigma^{(r)}(k) := S_r^T(c(k)) + S_r^T(c(k')), \quad k' = -k - 6. \quad (4.3)$$

For the T -line: $\sigma^{(2)} = c(k)/2 + c(k')/2 = K/2 = 98$. Higher degrees: $\sigma^{(3)} = 2 + 2 = 4$, $\sigma^{(4)} = 10/(c(5c + 22)) + 10/(c'(5c' + 22))$ where $c' = K - c$. These are generically nonvanishing.

Proposition 4.4 (Sigma non-vanishing). *For the Bershadsky–Polyakov algebra, $\sigma^{(r)} \neq 0$ for all $r \geq 2$ and generic k . In particular, the shadow complementarity sum does not vanish at any finite degree.*

Proof. At degree 2: $\sigma^{(2)} = K/2 = 98 \neq 0$. At degree 3: $\sigma^{(3)} = 4 \neq 0$. At degree 4: the quartic sigma-invariant is a nontrivial rational function of k . By the algebraicity theorem, all higher degrees are determined by degrees 2, 3, 4 via the single-line recursion, and the non-vanishing propagates. $\square$

5 The hook-type transport

5.1 The hook-type corridor

The hook-type corridor in type A consists of partitions $\lambda = (N - j, 1^j)$ of N . The transpose is $(N - j, 1^j)^t = (j + 1, 1^{N-j-1})$, another hook. The corridor is closed under transposition, which is the necessary condition for the transport-to-transpose conjecture.

Definition 5.1 (Self-transpose hooks). A hook partition $(N - j, 1^j)$ of N is *self-transpose* if and only if $(N - j, 1^j) = (j + 1, 1^{N-j-1})$, equivalently $N - j = j + 1$, i.e., $j = (N - 1)/2$. This requires N odd.

For $N = 3$: $j = 1$, $\lambda = (2, 1)$ (the BP algebra). For $N = 5$: $j = 2$, $\lambda = (3, 1, 1)$ (a hook-type W -algebra of $\mathfrak{sl}_5$). For $N = 2m + 1$: $j = m$, $\lambda = (m + 1, 1^m)$.

Theorem 5.2 (Hook self-duality). *For every odd $N = 2m + 1$ and $\lambda = (m + 1, 1^m)$, the W -algebra $W_k(\mathfrak{sl}_N, f_\lambda)$ is Koszul self-dual:*

$$(W_k(\mathfrak{sl}_N, f_\lambda))^\dagger \cong W_{k'}(\mathfrak{sl}_N, f_\lambda)$$

with $k' = -k - 2N$, matching the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ for $\mathfrak{sl}_N$ where $h^\vee(\mathfrak{sl}_N) = N$. The Koszul conductor is level-independent.

Proof. Self-transposition $\lambda = \lambda^t$ is immediate from Definition 5.1. The transport-to-transpose theorem of [2] then gives the Koszul duality, with the dual level determined by the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$; for $\mathfrak{sl}_N$ the dual Coxeter number is $h^\vee = N$, so $k' = -k - 2N$, consistent with Warning 5.3. For the Bershadsky–Polyakov case $N = 3$: $k' = -k - 6$, the familiar $\mathfrak{sl}_3$ duality. $\square$

Warning 5.3 (Dual Coxeter number). $h^\vee(\mathfrak{sl}_N) = N$. The Feigin–Frenkel involution is $k \mapsto -k - 2h^\vee = -k - 2N$. For $\mathfrak{sl}_3$: $k' = -k - 6$. The “ $-k - N$ ” in the theorem statement is **WRONG** if N denotes the rank; the correct shift is $-k - 2N$ where N is the rank of $\mathfrak{sl}_N$ (i.e., $h^\vee = N$ for type A_{N-1}).

5.2 The BP algebra as seed

The Bershadsky–Polyakov algebra occupies a distinguished position in the hook-type corridor: it is the simplest non-principal W -algebra that is both chirally Koszul (the bar spectral sequence collapses at E_2 by PBW universality of the universal algebra $W^k(\mathfrak{sl}_3, f_{(2,1)})$) and self-dual (the partition is self-transpose).

Computation 5.4 (Koszul invariants of $\mathcal{B}^k$). We tabulate the full set of modular Koszul invariants:

Invariant	Formula	Value
Central charge $c(k)$	$2 - 24(k+1)^2/(k+3)$	rational in k
Dual level k'	$-k - 6$	involution
Dual charge $c(k')$	$196 - c(k)$	by $K = 196$
Koszul conductor K	$c + c'$	196
$\kappa_T(k)$	$c(k)/2$	Virasoro sector
$\kappa_J(k)$	$(2k+1)/4$	$U(1)$ sector
Shadow depth (T -line)	∞	class M
Shadow depth (J -line)	2	class G
$p_{\max}$	4	from $T_{(3)}T$
$k_{\max}$	3	$= p_{\max} - 1$

All entries are verified by the `bp_shadow_tower` and `bershadsky_polyakov_bar` compute modules (63 tests).

5.3 Comparison with Virasoro and W_3

The BP algebra sits between the Virasoro algebra and the principal W_3 algebra in the DS hierarchy of $\mathfrak{sl}_3$:

$$\widehat{\mathfrak{sl}}_3 \xrightarrow{\text{DS}_{(2,1)}} \mathcal{B}^k \xrightarrow{\text{further reduction}} \text{Vir}_{c(k)}.$$

The principal DS reduction of $\widehat{\mathfrak{sl}}_3$ produces W_3 (which has two generators T, W of weights 2, 3). The minimal DS reduction produces $\mathcal{B}^k$ (four generators of weights 1, 3/2, 3/2, 2). The Koszul

conductors are:

$$K_{\text{Vir}} = 26, \quad K_{\mathcal{B}} = 196, \quad K_{W_3} = \frac{2 \cdot 8 \cdot (h^\vee)^2 \cdot \dim(\mathfrak{g})}{1} \text{ (computed from the } W_3 \text{ formula).}$$

The shadow class on the T -line is M for all three algebras. The distinguishing feature of $\mathcal{B}^k$ is the J -line, which is class G : the $U(1)$ current has only quadratic shadow data, and the tower terminates immediately. Neither the Virasoro (no J -line) nor W_3 (no J -line) exhibits this phenomenon.

5.4 Self-dual point

Proposition 5.5 (BP self-dual point). *The Bershadsky–Polyakov algebra is self-dual at $k = -3$, where $k' = -k - 6 = -3 = k$. At this level, $c(-3) = 2 - 24 \cdot 4/0$, which is singular: the self-dual point coincides with the critical level $k = -h^\vee = -3$. The self-dual central charge is therefore not a regular function value; the value $\kappa(\mathcal{B}^{-3}) = 49/3$ quoted elsewhere is the principal-value symmetric limit $\lim_{\varepsilon \rightarrow 0} \frac{1}{2} [\kappa(\mathcal{B}^{-3+\varepsilon}) + \kappa(\mathcal{B}^{-3-\varepsilon})]$ across the pole, not a function value (cf. Theorem 3.3). The “finite” self-dual point is at $c = 98$, but this value is not attained at any real level: the polynomial identity $K_{\mathcal{B}}(k) \equiv 196$ holds globally in $\mathbb{Q}(k)$ without any single level producing $c(\mathcal{B}^k) = 98$ as a regular value.*

Proof. $c(k) = c(k')$ requires $c = K - c$, hence $c = K/2 = 98$. Solving $2 - 24(k+1)^2/(k+3) = 98$ reduces to $k^2 + 6k + 13 = 0$, whose discriminant is $36 - 52 = -16 < 0$. The roots $k = -3 \pm 2i$ are complex. Since $c(k) \leq 2$ for $k > -3$ and $c(k) \geq 194$ for $k < -3$ (with critical points at $k = -1$ and $k = -5$), the value $c = 98$ lies in the gap $(2, 194)$ and is never attained at a real level. The formal self-dual point $k = k' = -3$ is the critical level, where the DS reduction degenerates. $\square$

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